Chapter 1. Introduction and Elementary Concepts

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1 Collection of Survey Data

1.1 Primary and secondary data

Definition 1.1

- **Primary data**: data collected by the investigator directly from the source.
- **Secondary data**: data that has already been collected by some agency or individual and is available in published or unpublished records.

Primary data fit the study's target; secondary data are cheaper but collected for another purpose. Primary data can be collected by

- direct personal interviews, or interviews by enumerators,
- questionnaires sent by mail,
- telephone interviews.

1.2 Framing a questionnaire

The person framing the questionnaire should know the field of inquiry in detail. Points to keep in mind:

- The surveyor introduces themselves and states the objective of the survey.

- Questions are clear, unambiguous and to the point; leading questions are avoided.
- Questions affecting the prestige or sentiments of respondents, and questions requiring calculations, are avoided.
- Precise instructions for filling in the questionnaire, including units of measurement, are given.
- The questionnaire is concise: keep the number of questions close to the number of quantities to be estimated.
- The layout is visually appealing.

1.3 Technical terms

Definition 1.2

- An **element** is a unit for which information is sought.
- The **population** (universe) is the aggregate of elements about which inference is to be made.
- **Sampling units** are non-overlapping collections of population elements.
- The **sampling frame** is a list of all sampling units in the population.
- A **sample** is a subset of the population selected from the frame to draw inferences about population characteristics.

Observing every unit is a **complete enumeration** or **census**. A census is often impossible, and a sample survey is faster, more detailed, cheaper and often more accurate (trained staff and follow-up reduce non-sampling errors).

1.4 Probability and non-probability sampling

The method used to select the sample is the **sampling procedure**.

- **Probability sampling**: units are selected by a probability mechanism. Only these procedures are considered in this course.
- **Non-probability sampling**: units are selected without a probability mechanism (convenience, quota, judgement); such procedures are subject to human bias.

1.5 With and without replacement

Definition 1.3

- **With replacement (WR)**: each drawn unit is returned before the next draw; with order, N^n possible samples.
- **Without replacement (WOR)**: the selected unit is not returned before the next draw. There are $\binom{N}{n}$ possible (unordered) samples.

Example 1.4: Birth weights

Four children A, B, C, D have birth weights $5.5, 8.0, 6.5, 7.0$ pounds. The WR samples of size 2 (ordered) are the $4^2 = 16$ pairs

$$(A,A),(A,B),(A,C),(A,D),(B,A),\ldots,(D,D),$$

whereas WOR gives the $\binom{4}{2} = 6$ samples $\{A,B\},\{A,C\},\{A,D\},\{B,C\},\{B,D\},\{C,D\}$.

In R: `sample(n)` is a random permutation of $1,\ldots,n$; `sample(x)` permutes x; `sample(x, n)` draws n items WOR; `sample(x, n, replace=T)` draws WR; `sample(x, replace=T)` is a bootstrap sample.

1.6 Planning and execution of a sample survey

Objectives → population to be studied → sampling unit → sampling frame → sample selection (sample size and procedure) → method of collecting information → pilot survey → handling of nonresponse → organization of fieldwork → analysis of data and report.

2 Elementary Concepts

2.1 Expectation, variance, covariance

If x takes values $x_1,\ldots,x_k$ with probabilities $p_1,\ldots,p_k$,

$$\mathrm{E}(x) = \sum_{i=1}^{k} p_i x_i, \qquad \mathrm{Var}(x) = \mathrm{E}[x-\mathrm{E}(x)]^2 = \mathrm{E}(x^2) - [\mathrm{E}(x)]^2,$$

$$\mathrm{Cov}(x,y) = \mathrm{E}[(x-\mathrm{E}x)(y-\mathrm{E}y)] = \mathrm{E}(xy) - \mathrm{E}(x)\mathrm{E}(y),$$

$$\mathrm{Var}\Big(\sum_{i=1}^{k} a_i x_i\Big) = \sum_{i=1}^{k} a_i^2 \mathrm{Var}(x_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(x_i,x_j).$$

2.2 Parameters and statistics

Definition 1.5

A **parameter** is a real function of all population values, a **statistic** a real function of the sample values; a statistic estimating a parameter is an **estimator**.

For population values $Y_1,\ldots,Y_N$ and sample values $y_1,\ldots,y_n$:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} Y_i, \quad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(Y_i-\mu)^2, \quad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i-\mu)^2 = \frac{N}{N-1}\sigma^2,$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2.$$

The finite-population variance S^2 (divisor $N-1$) is the natural target of s^2 under SRS without replacement (Ch. 2), and $(N-1)S^2 = N\sigma^2$.

2.3 Bias, variance and MSE

Definition 1.6

- $\hat{\theta}$ is **unbiased** for θ if $\mathrm{E}(\hat{\theta}) = \theta$. Otherwise its **bias** is $B(\hat{\theta}) = \mathrm{E}(\hat{\theta}) - \theta$ and its **relative bias** is $RB(\hat{\theta}) = B(\hat{\theta})/\theta$.
- Sampling variance $\mathrm{Var}(\hat{\theta}) = \mathrm{E}[\hat{\theta} - \mathrm{E}\hat{\theta}]^2$; standard error $SE(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$.
- Mean squared error $\mathrm{MSE}(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta)^2$.
- The **relative efficiency** of $\hat{\theta}_2$ with respect to $\hat{\theta}_1$ is $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{MSE}(\hat{\theta}_1)/\mathrm{MSE}(\hat{\theta}_2)$; if it exceeds 1, $\hat{\theta}_2$ is more efficient.

Theorem 1.7: Bias–variance decomposition

$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$.

Proof.

$$\begin{aligned}\mathrm{E}(\hat{\theta} - \theta)^2 &= \mathrm{E}\big[(\hat{\theta} - \mathrm{E}\hat{\theta}) + (\mathrm{E}\hat{\theta} - \theta)\big]^2 \\ &= \mathrm{E}(\hat{\theta} - \mathrm{E}\hat{\theta})^2 + (\mathrm{E}\hat{\theta} - \theta)^2 + 2(\mathrm{E}\hat{\theta} - \theta)\,\mathrm{E}(\hat{\theta} - \mathrm{E}\hat{\theta}),\end{aligned}$$

and the last expectation is zero. ■

2.4 Confidence intervals and the bound on the error

For a desired probability $1 - \alpha$ we look for a bound B with

$$P(|\hat{\theta} - \theta| \le B) = 1 - \alpha \iff P(\hat{\theta} - B \le \theta \le \hat{\theta} + B) = 1 - \alpha,$$

so $(\hat{\theta} - B, \hat{\theta} + B)$ is a confidence interval with confidence coefficient $1 - \alpha$.

- **Normal case**: for (approximately) normal unbiased $\hat{\theta}$, $B = z_{\alpha/2}\sigma_{\hat{\theta}}$; at 95%, $z_{0.025} \approx 2$ gives the "2-SD bound" $B = 2\sigma_{\hat{\theta}}$ used throughout.
- **Any distribution**: by Chebyshev's inequality $P(|\hat{\theta} - \theta| \le 2\sigma_{\hat{\theta}}) \ge 1 - \frac{1}{4} = 0.75$.

2.5 Sample size determination

The required precision $P(|\hat{\theta} - \theta| \le B) = 1 - \alpha$ is set by the investigator, with **tolerable error** (margin of error) B; for proportions in surveys of people, typically $B = 0.03$, $\alpha = 0.05$.

Example 1.8: Opinion polls

With the normal approximation and $p = 1/2$ (the worst case),

$$B = 1.96\sqrt{\frac{p(1-p)}{n}} \le 1.96\sqrt{\frac{1}{4n}}.$$

Setting $B = 0.03$ gives $n = (1.96)^2/(4{\cdot}0.03^2) = 1067.1$, so about 1,068 respondents; this is why national polls typically report a margin of error of $\pm 3\%$ with about a thousand respondents.

3 Selected exercises

Motivation. The 2-SD guarantee, an exact sampling distribution by enumeration, and a biased estimator beating an unbiased one in MSE.

Exercise 1.1 (Chebyshev guarantee for the bound on the error)

Let $\hat{\theta}$ be an estimator of θ with finite variance, and let $k > 0$.
(a) If $\hat{\theta}$ is unbiased with standard error $\sigma_{\hat{\theta}} > 0$, show that $P(|\hat{\theta} - \theta| \le k\sigma_{\hat{\theta}}) \ge 1 - 1/k^2$.
(b) Without assuming unbiasedness, show that $P(|\hat{\theta} - \theta| \le k\sqrt{\mathrm{MSE}(\hat{\theta})}) \ge 1 - 1/k^2$ whenever $\mathrm{MSE}(\hat{\theta}) > 0$, and explain why the 2-SD bound of a biased estimator should be built from $\sqrt{\mathrm{MSE}}$ rather than from the standard error.

Solution. [Supplement] Both parts follow from one inequality.

Step 1. (Markov's inequality.) For a random variable $U \ge 0$ and $a > 0$, $a\,\mathbf{1}\{U \ge a\} \le U$ pointwise; taking expectations gives $P(U \ge a) \le \mathrm{E}(U)/a$.

Step 2. Apply Step 1 to $U = (\hat{\theta} - \theta)^2$ and $a = k^2\mathrm{E}(\hat{\theta} - \theta)^2 = k^2\mathrm{MSE}(\hat{\theta}) > 0$:

$$P(|\hat{\theta} - \theta| > k\sqrt{\mathrm{MSE}(\hat{\theta})}) \le P((\hat{\theta} - \theta)^2 \ge k^2\mathrm{MSE}(\hat{\theta})) \le \frac{\mathrm{MSE}(\hat{\theta})}{k^2\mathrm{MSE}(\hat{\theta})} = \frac{1}{k^2}.$$

Taking complements proves (b).

(a) If $\mathrm{E}\hat{\theta} = \theta$, $\mathrm{MSE}(\hat{\theta}) = \sigma^2_{\hat{\theta}}$, so (a) is a special case of (b); $k = 2$ gives at least $3/4$.

For biased $\hat{\theta}$, $\mathrm{MSE} = \sigma^2_{\hat{\theta}} + B^2$, and $|\hat{\theta} - \theta| \le 2\sigma_{\hat{\theta}}$ may have probability below $1/2$ (normal $\hat{\theta}$, $|B| > 2\sigma_{\hat{\theta}}$); use $\sqrt{\mathrm{MSE}}$. ■

Exercise 1.2 (Birth weights: exact sampling distribution of $\bar{y}$)

For the birth weights $5.5, 8.0, 6.5, 7.0$ (example on WR and WOR sampling), let $\bar{y}$ be the mean of a sample of size $n = 2$.
(a) Under WOR, list the six sample means and compute $\mathrm{E}(\bar{y})$ and $\mathrm{Var}(\bar{y})$ by enumeration.
(b) Under WR (ordered pairs, each with probability $1/16$), compute $\mathrm{Var}(\bar{y})$ and the ratio of the two variances. Compare with σ^2/n and with $\frac{N-n}{N-1}\frac{\sigma^2}{n}$.

Solution. The population mean and variance are $\mu = 27/4 = 6.75$ and

$$\sigma^2 = \tfrac{1}{4}[(-1.25)^2 + 1.25^2 + (-0.25)^2 + 0.25^2] = \tfrac{3.25}{4} = 0.8125.$$

(a) The six samples, each with probability $1/6$, have means

$\{A,B\}$	$\{A,C\}$	$\{A,D\}$	$\{B,C\}$	$\{B,D\}$	$\{C,D\}$
6.75	6.00	6.25	7.25	7.50	6.75

Their sum is 40.5, so $\mathrm{E}(\bar{y}) = 40.5/6 = 6.75 = \mu$. The squared deviations from 6.75 are 0, 0.5625, 0.25, 0.25, 0.5625, 0, so

$$\mathrm{Var}(\bar{y}) = \frac{1.625}{6} = 0.2708.$$

(b) Under WR the draws are independent and uniform, so $\mathrm{Var}(\bar{y}) = \sigma^2/2 = 0.40625$ (as enumeration of the 16 ordered pairs confirms). The ratio is

$$\frac{0.40625}{0.2708} = 1.5 = \frac{N-1}{N-n} = \frac{3}{2}.$$

So WR gives σ^2/n and WOR $\frac{N-n}{N-1}\frac{\sigma^2}{n} = \frac{2}{3}\cdot 0.40625$ (Ch. 2): WOR is 50% more efficient here, observing half the population. ■

Exercise 1.3 (Shrinking the sample mean)

Let $y_1, \ldots, y_n$ be iid with mean $\mu \neq 0$ and variance $\sigma^2 > 0$, and consider $\hat{\theta}_c = c\,\bar{y}$ for a constant c.
(a) Find $\mathrm{MSE}(\hat{\theta}_c)$ as a function of c.
(b) Find the minimizing c^* and show that $\mathrm{MSE}(\hat{\theta}_{c^*}) < \mathrm{MSE}(\bar{y})$.
(c) Compute $\mathrm{RE}(\bar{y}, \hat{\theta}_{c^*})$ and discuss why $\hat{\theta}_{c^*}$ is not used in practice.

Solution. [Supplement]

(a) $\mathrm{E}(\hat{\theta}_c) = c\mu$ and $\mathrm{Var}(\hat{\theta}_c) = c^2\sigma^2/n$, so by the bias–variance decomposition

$$\mathrm{MSE}(c) = c^2\frac{\sigma^2}{n} + (c-1)^2\mu^2.$$

(b) $\mathrm{MSE}(c)$ is a quadratic in c with leading coefficient $\sigma^2/n + \mu^2 > 0$, and $\mathrm{MSE}'(c) = 2c\sigma^2/n + 2(c-1)\mu^2 = 0$ gives

$$c^* = \frac{\mu^2}{\mu^2 + \sigma^2/n} \in (0, 1).$$

Substituting, with $v = \sigma^2/n$,

$$\mathrm{MSE}(c^*) = \frac{\mu^4 v + v^2\mu^2}{(\mu^2+v)^2} = \frac{\mu^2 v}{\mu^2 + v} < v = \mathrm{MSE}(\bar{y}),$$

the inequality being strict because $v > 0$.

(c) $\mathrm{RE}(\bar{y}, \hat{\theta}_{c^*}) = (\mu^2 + v)/\mu^2 = 1 + \sigma^2/(n\mu^2) > 1$: shrinkage wins, especially when σ/μ is large relative to $\sqrt{n}$. But c^* involves the unknown σ^2/μ^2, and a plug-in c loses the guarantee of (b). Unbiasedness is a convention; MSE is the criterion. ■